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Primary controller applied to a primary side of a power converter and operation method thereof

Abstract

A primary controller applied to a primary side of a power converter includes a current peak upper limit adjustment circuit and a gate control signal generation circuit. The current peak upper limit adjustment circuit increases a current peak upper limit of the primary side of the power converter when a gate control signal is enabled at a second valley of a voltage, and reduces the current peak upper limit when the gate control signal is enabled at an Nth valley of the voltage, wherein N is a positive integer. The gate control signal generation circuit is coupled to the current peak upper limit adjustment circuit, wherein the gate control signal generation circuit enables the gate control signal, disables the gate control signal according to the current peak upper limit, and the gate control signal is used for turning on a power switch of the primary side of the power converter.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a primary controller applied for a primary side of a power converter and an operational method thereof, and particularly to a primary controller and an operational method thereof that can regulate a current peak upper limit according to a voltage corresponding to a drain voltage of a synchronous switch of a second side of the power converter so that conduction loss of the power converter is significantly reduced and an output voltage the power converter can be regulated under a condition of the power converter operating in a discontinuous-conduction mode (DCM) and having a constant ON time (COT) control.

2. Description of the Prior Art

(2) In the prior art, when an output voltage of a power converter with a constant ON time (COT) control is less than a reference voltage, a pulse width modulation signal inputted into the power converter can be enabled on the constant ON time. If the power converter operates in a discontinuous-conduction mode (DCM), the constant ON time corresponds to a constant current peak upper limit. That is, the power converter operates in pulse frequency modulation. Meanwhile, the output voltage is regulated by a frequency of the pulse width modulation signal.

(3) The pulse frequency modulation can minimize switching loss of the power converter when a load of the power converter is light. However, when the load of the power converter is heavy and an alternating voltage inputted into the power converter is low line, because duty utility of the pulse width modulation signal is lower and a root mean square current of the power converter is larger, conduction loss of the power converter becomes very serious. In addition, under a condition of the pulse frequency modulation, if the current peak upper limit is fixed to make the frequency of the pulse width modulation signal fixed, the output voltage cannot be regulated when the power

converter operates in a boundary current mode (BCM).

(4) Therefore, how to solve disadvantages of the output voltage of the power converter not capable of being regulated and the power converter having more serious conduction loss will be an important issue for a designer of the power converter.

SUMMARY OF THE INVENTION

(5) An embodiment of the present invention provides a primary controller applied for a primary side of a power converter. The primary controller includes a current peak upper limit adjustment circuit and a gate control signal generation circuit. The current peak upper limit adjustment circuit is used for increasing a current peak upper limit of the primary side of the power converter when a gate control signal is enabled at a second valley of a voltage, and reducing the current peak upper limit when the gate control signal is enabled at an N^{th} valley of the voltage, wherein N is a positive integer. The gate control signal generation circuit is coupled to the current peak upper limit adjustment circuit, wherein the gate control signal generation circuit enables the gate control signal, disables the gate control signal according to the current peak upper limit, and the gate control signal is used for turning on a power switch of the primary side of the power converter.

(6) Another embodiment of the present invention provides an operational method of a primary controller applied to a primary side of a power converter, wherein the primary controller includes a current peak upper limit adjustment circuit and a gate control signal generation circuit. The operational method includes the current peak upper limit adjustment circuit increasing a current peak upper limit of a primary side of the power converter when a gate control signal is enabled at a second valley of a voltage, and reducing the current peak upper limit when the gate control signal is enabled at an N^{th} valley of the voltage, wherein N is a positive integer; and the gate control signal generation circuit enabling the gate control signal, and disabling the gate control signal according to the current peak upper limit, wherein the gate control signal is used for turning on a power switch of the primary side of the power converter.

(7) The present invention provides a primary controller applied for a primary side of a power converter and an operational method thereof. The primary controller and the operational method can receive a voltage corresponding to a drain voltage of a synchronous switch of a second side of the power converter through a primary side auxiliary winding and a second side winding of the power converter and regulate a current peak upper limit of a primary side of the power converter accordingly under a condition that the power converter cannot detect an output voltage of the second side of the power converter to regulate the current peak upper limit and operates in a discontinuous-conduction mode. Therefore, compared to the prior art, the present invention has advantages as follows: first, because the primary controller can regulate the current peak upper limit to make the power converter operate approximately in the boundary current mode, duty utility of a gate control signal generated by the primary controller is higher, resulting in conduction loss of the power converter being significantly reduced; second, because the primary controller can regulate the current peak upper limit, the output voltage can be regulated.

(8) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating a primary controller applied to a primary side of a power converter according to a first embodiment of the present invention.

(2) FIG. 2 is a diagram illustrating waveforms of a drain voltage of the synchronous switch of the secondary side of the power converter and the gate control signal.

(3) FIG. 3 is a flowchart illustrating an operational method of the primary controller applied to the primary side of the power converter according to a second embodiment of the present invention.

DETAILED DESCRIPTION

(4) Please refer to FIG. 1. FIG. 1 is a diagram illustrating a primary controller **200** applied to a primary side PRI of a power converter **100** according to a first embodiment of the present invention, wherein as shown FIG. 1, the primary controller **200** at least includes a current peak upper limit adjustment circuit **202** and a gate control signal generation circuit **204**, and the current peak upper limit adjustment circuit **202** includes a counter **2022**, a digital to analog converter (DAC) **2024**, and an adder **2026**. The gate control signal generation circuit **204** is coupled to the current peak upper limit adjustment circuit **202**, the digital to analog converter **2024** is coupled to the counter **2022**, and the adder **2026** is coupled to the digital to analog converter **2024**. In addition, the power converter **100** is a flyback power converter, and the power converter **100** operates in a discontinuous-conduction mode (DCM). In addition, the primary controller **200** is applied for a condition of the primary side PRI of the power converter **100** having a constant ON time (COT) control, and an output voltage VOUT of a secondary side SEC of the power converter **100** is regulated by a switching frequency of a gate control signal GCS which is used for turning on a power switch **102** of the primary side PRI of the power converter **100**. In addition, for the sake of simplicity, the primary side PRI of the power converter **100** and the secondary side SEC of the power converter **100** only show elements corresponding to the present invention. In addition, as shown in FIG. 1, operational principles of how a secondary controller **300** applied to the secondary side SEC of the power converter **100** controls a synchronous switch **104** according to the output voltage VOUT/an output current IOOUT of the secondary side SEC of the power converter **100** can be referred to Taiwan patent application No. I681615, so further description thereof is omitted for simplicity. In addition, an input voltage VIN of the primary side PRI of the power converter **100** is generated by rectifying an alternating voltage through a bridge rectifier (not shown in FIG. 1).

(5) After the primary side PRI of the power converter **100** is turned off (that is, after the gate control signal GCS for turning on the power switch **102** of the primary side PRI of the power converter **100** is disabled), waveforms of a drain voltage VDS of the synchronous switch **104** of the secondary side SEC of the power converter **100** and the gate control signal GCS can be referred to FIG. 2, wherein the drain voltage VDS can be coupled to a primary side auxiliary winding **106** of the power converter **100** to make the primary side auxiliary winding **106** generate a voltage VC, a waveform of the voltage VC is similar to the waveform of the drain voltage VDS, and the gate control signal GCS is a pulse width modulation signal. Therefore, when the output voltage VOUT of the secondary side SEC of the power converter **100** is less than a reference voltage VREF, because the power converter **100** operates in the discontinuous-conduction mode, the gate control signal generation circuit **204** enables the gate control signal GCS to turn on the power switch **102** at most at a time T1 corresponding to a second valley (a dashed-line circle A shown in FIG. 2) of the voltage VC, wherein the gate control signal generation circuit **204** enabling the gate control signal GCS to turn on the power switch **102** at the time T1 corresponding to the second valley (the dashed-line circle A shown in FIG. 2) of the voltage VC can make the power converter **100** operate approximately in a boundary current mode (BCM). In addition, because the waveform of the voltage VC is similar to the waveform of the drain voltage VDS, the second valley of the voltage VC corresponds to a second valley of the drain voltage VDS. Meanwhile, the counter **2022** coupled to the gate control signal generation circuit **204** increases a current count stored in the counter **2022** to a first count FC (e.g. the counter **2022** adds one to the current count (e.g. 0) to generate the first count FC (i.e. 1)). Then, the digital to analog converter **2024** can generate one unit current variation (wherein the unit current variation corresponds to a difference 1 between the first count FC (1) and the current count (0)) according to the first count FC (1) and the current count (0). Finally, the adder **2026** makes a current peak upper limit CPL (corresponding to the current count (i.e. 0)) be increased the unit current variation to become a first new current peak upper limit FNCPL

(corresponding to the first count FC (i.e. 1)) according to the unit current variation. After the gate control signal generation circuit **204** receives the first new current peak upper limit FNCPL, the gate control signal generation circuit **204** can adjust a voltage upper limit. Then, the gate control signal generation circuit **204** will disable the gate control signal GCS in a next period of the power converter **100** according to the voltage upper limit and a detection voltage VD, wherein because the first new current peak upper limit FNCPL is greater than the current peak upper limit CPL, the voltage upper limit is also increased, resulting in an ON time of the gate control signal GCS being also increased (i.e. as shown in FIG. 2, an ON time T2-T22 (corresponding to the first count FC (i.e. 1)) corresponding to the next period of the gate control signal GCS is greater than an ON time T1-T11 (corresponding to the current count (i.e. 0)) corresponding to a current period of the gate control signal GCS). Thus, the output voltage VOUT is also increased accordingly. In addition, as shown in FIG. 1, the detection voltage VD is determined by a primary side current IPRI of the primary side PRI of the power converter **100** and a resistor **108**. In addition, as shown in FIG. 2, at the time T2 and the time T3, operational principles of the gate control signal generation circuit **204** enabling the gate control signal GCS corresponding to the second valley of the voltage VC can be referred to operational principles of the gate control signal generation circuit **204** enabling the gate control signal GCS corresponding to the second valley of the voltage VC at the time T1, so further description thereof is omitted for simplicity. In addition, after the time T3, the current count stored in the counter **2022** is 3. In addition, one of ordinary skilled in the art should know an auxiliary voltage VNAUX of the primary side auxiliary winding **106** can be used for generating a supply voltage VCC of the primary controller **200**.

(6) When the output voltage VOUT of the secondary side SEC of the power converter **100** is not less than the reference voltage VREF, the gate control signal generation circuit **204** enables the gate control signal GCS to turn on the power switch **102** at a time T4 which corresponds to an N^{sup}.th valley (e.g. the fourth valley at a dashed-line circle B in FIG. 2) of the voltage VC, wherein N is a positive integer. In addition, because the waveform of the voltage VC is similar to the waveform of the drain voltage VDS, the fourth valley of the voltage VC corresponds to a fourth valley of the drain voltage VDS. Meanwhile, the current count stored in the counter **2022** will become a second count SC by the counter **2022** subtracting N-M from the current count stored in the counter **2022**, wherein M is less than N and M is a positive integer greater than 2 (e.g. the current count (i.e. 3) will become the second count SC (i.e. 1) by the counter **2022** subtracting (4-2) from the current count (i.e. 3), and N-M is in proportion to a dead time of the power converter **100**. Then, the digital to analog converter **2024** can generate two unit current variations (wherein the two unit current variations correspond to a difference 2 between the second count SC (1) and the current count (3)) according to the second count SC (1) and the current count (3). Finally, the adder **2026** makes the current peak upper limit CPL be reduced the two unit current variations to become a second new current peak upper limit SNCPL (corresponding to the second count SC (i.e. 1)) according to the two unit current variations. After the gate control signal generation circuit **204** receives the second new current peak upper limit SNCPL, the gate control signal generation circuit **204** can adjust the voltage upper limit. Then, the gate control signal generation circuit **204** will disable the gate control signal GCS in a next period of the power converter **100** according to the voltage upper limit and the detection voltage VD, wherein because the second new current peak upper limit SNCPL is less than the current peak upper limit CPL, the voltage upper limit is also reduced, resulting in the ON time of the gate control signal GCS being also reduced (i.e. as shown in FIG. 2, an ON time T5-T55 (corresponding to the second count SC (i.e. 1)) corresponding to the next period of the gate control signal GCS is less than an ON time T4-T44 (corresponding to the current count (i.e. 3))) corresponding to the current period of the gate control signal GCS. Thus, the output voltage VOUT is also reduced accordingly. In addition, because an ON time T5-T55 of the gate control signal GCS corresponds to the second count SC (i.e. 1), the ON time of the gate control signal GCS will be equal to the ON time T2-T22 (corresponding to the first count FC (i.e. 1) of the gate control

signal GCS).

(7) In addition, in another embodiment of the present invention, the power converter **100** can be an active-clamp flyback (ACF) power converter) or an asymmetric half-bridge (AHB) flyback power converter.

(8) In addition, please refer to FIG. 1, FIG. 2, FIG. 3. FIG. 3 is a flowchart illustrating an operational method of the primary controller **200** applied to the primary side PRI of the power converter **100** according to a second embodiment of the present invention. Detailed steps of the operational method shown in FIG. 3 are as follows: Step **300**: Start. Step **302**: The power converter **100** is powered on. Step **304**: When the gate control signal GCS is enabled at the second valley of the voltage VC, go to Step **306**; when the gate control signal GCS is enabled at the N.sup.th valley of the voltage VC, go to Step **308**. Step **306**: The current peak upper limit adjustment circuit **202** increase the current peak upper limit CPL of the primary side PRI of the power converter **100** to generate the first new current peak upper limit FNCPL, go to Step **310**. Step **308**: The current peak upper limit adjustment circuit **202** reduces the current peak upper limit CPL to generate the second new current peak upper limit SNCPL, go to Step **312**. Step **310**: The gate control signal generation circuit **204** regulates the gate control signal GCS according to the first new current peak upper limit FNCPL, go to Step **304**. Step **312**: The gate control signal generation circuit **204** regulates the gate control signal GCS according to the second new current peak upper limit SNCPL, go to Step **304**.

(9) In Step **304**, when the output voltage VOUT of the secondary side SEC of the power converter **100** is less than the reference voltage VREF, because the power converter **100** operates in the discontinuous-conduction mode, the gate control signal generation circuit **204** enables the gate control signal GCS to turn on the power switch **102** at most at the time T1 corresponding to the second valley (the dashed-line circle A shown in FIG. 2) of the voltage VC, wherein the gate control signal generation circuit **204** enabling the gate control signal GCS to turn on the power switch **102** at the time T1 corresponding to the second valley (the dashed-line circle A shown in FIG. 2) of the voltage VC can make the power converter **100** operate approximately in a boundary current mode (BCM).

(10) In Step **306**, meanwhile, the counter **2022** increases the current count stored in the counter **2022** to the first count FC (e.g. the counter **2022** adds one to the current count (e.g. 0) to generate the first count FC (i.e. 1)). Then, the digital to analog converter **2024** can generate one unit current variation (wherein the unit current variation corresponds to a difference 1 between the first count FC (1) and the current count (0)) according to the first count FC (1) and the current count (0). Finally, the adder **2026** makes the current peak upper limit CPL be increased the unit current variation to become the first new current peak upper limit FNCPL according to the unit current variation.

(11) In Step **310**, after the gate control signal generation circuit **204** receives the first new current peak upper limit FNCPL, the gate control signal generation circuit **204** can adjust the voltage upper limit. Then, the gate control signal generation circuit **204** will disable the gate control signal GCS in a next period of the power converter **100** according to the voltage upper limit and the detection voltage VD, wherein because the first new current peak upper limit FNCPL is greater than the current peak upper limit CPL, the voltage upper limit is also increased, resulting in an ON time of the gate control signal GCS being also increased. Thus, the output voltage VOUT is also increased accordingly.

(12) In addition, in Step **304**, when the output voltage VOUT of the secondary side SEC of the power converter **100** is not less than the reference voltage VREF, the gate control signal generation circuit **204** enables the gate control signal GCS to turn on the power switch **102** at the time T4 which corresponds to the N.sup.th valley (e.g. the fourth valley at the dashed-line circle B in FIG. 2) of the voltage VC, wherein N is a positive integer.

(13) In Step **308**, meanwhile, the current count stored in the counter **2022** will become the second count SC by the counter **2022** subtracting N-M from the current count stored in the counter **2022**,

wherein M less than N and M is a positive integer not less than 2 (e.g. the counter **2022** (i.e. 3) will become the second count SC (i.e. 1) by the counter **2022** subtracting $(4-2)$ from the current count (i.e. 3)). Then, the digital to analog converter **2024** can generate the two unit current variations according to the second count SC (1) and the current count (3). Finally, the adder **2026** makes the current peak upper limit CPL be reduced the two unit current variations to become the second new current peak upper limit SNCPL according to the two unit current variations.

(14) In Step **312**, after the gate control signal generation circuit **204** receives the second new current peak upper limit SNCPL, the gate control signal generation circuit **204** can adjust the voltage upper limit. Then, the gate control signal generation circuit **204** will disable the gate control signal GCS in a next period of the power converter **100** according to the voltage upper limit and the detection voltage VD, wherein because the second new current peak upper limit SNCPL is less than the current peak upper limit CPL, the voltage upper limit is also reduced, resulting in the ON time of the gate control signal GCS being also reduced. Thus, the output voltage VOUT is also reduced accordingly.

(15) To sum up, the primary controller and the operational method thereof can receive the voltage corresponding to the drain voltage through the primary side auxiliary winding and a second side winding of the power converter and regulate the current peak upper limit accordingly under a condition that the power converter cannot detect the output voltage to regulate the current peak upper limit and operates in the discontinuous-conduction mode. Therefore, compared to the prior art, the present invention has advantages as follows: first, because the primary controller can regulate the current peak upper limit to make the power converter operate approximately in the boundary current mode, duty utility of the gate control signal is higher, resulting in the conduction loss of the power converter being significantly reduced; second, because the primary controller can regulate the current peak upper limit, the output voltage can be regulated.

(16) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A primary controller applied for a primary side of a power converter, comprising: a current peak upper limit adjustment circuit for increasing one unit current variation to a current peak upper limit of the primary side of the power converter when a gate control signal is enabled at a second valley of a voltage, and subtracting $N-M$ unit current variations from the current peak upper limit when the gate control signal is enabled at an N .sup.th valley of the voltage, where N is a positive integer, M is less than N , and M is a positive integer greater than 2; and a gate control signal generation circuit coupled to the current peak upper limit adjustment circuit, wherein the gate control signal generation circuit enables the gate control signal, disables the gate control signal according to the current peak upper limit, and the gate control signal is used for turning on a power switch of the primary side of the power converter.

2. The primary controller of claim 1, wherein the current peak upper limit adjustment circuit comprises: a counter for increasing a current count stored in the counter to a first count when the gate control signal is enabled at the second valley of the voltage, and reducing the current count to a second count when the gate control signal is enabled at the N th valley of the voltage; a digital to analog converter (DAC) coupled to the counter, wherein the digital to analog converter generates the unit current variation according to the first count, and generates the $N-M$ unit current variations according to the second count; and an adder coupled to the digital to analog converter, wherein the digital to analog converter increases the current peak upper limit and reduces the current peak upper limit according to the unit current variation and the $N-M$ unit current variations,

respectively.

3. The primary controller of claim 1, wherein the power converter is a flyback power converter.

4. The primary controller of claim 1, wherein the power converter is an active-clamp flyback (ACF) power converter.

5. The primary controller of claim 1, wherein the power converter is an asymmetric half-bridge (AHB) flyback power converter.

6. The primary controller of claim 1, wherein the voltage relates to a drain voltage of a synchronous switch of a second side of the power converter.

7. The primary controller of claim 1, wherein the power converter operates in a discontinuous-conduction mode (DCM).

8. An operational method of a primary controller applied to a primary side of a power converter, wherein the primary controller comprises a current peak upper limit adjustment circuit and a gate control signal generation circuit, the operational method comprising: the current peak upper limit adjustment circuit increasing one unit current variation to a current peak upper limit of a primary side of the power converter when a gate control signal is enabled at a second valley of a voltage, and subtracting $N-M$ unit current variations from the current peak upper limit when the gate control signal is enabled at an N th valley of the voltage, wherein N is a positive integer, M is less than N , and M is a positive integer greater than 2; and the gate control signal generation circuit enabling the gate control signal, and disabling the gate control signal according to the current peak upper limit, wherein the gate control signal is used for turning on a power switch of the primary side of the power converter.

9. The operational method of claim 8, wherein the voltage relates to a drain voltage of a synchronous switch of a second side of the power converter.

10. The operational method of claim 8, wherein the power converter operates in a discontinuous-conduction mode.
